

Infrared linked illegal drone detection system

South Korea's Electronics and Telecommunications Research Institute (ETRI) has developed advanced technology to detect and track illegal drones, addressing security concerns like privacy invasion and terrorism. The system integrates a dual EO/IR (electro-optics/infrared) radar, co-developed



Public safety from illegal drones (Credit: ETRI)

with Hanwha Systems, to identify drones up to 5km away on the less-congested Ku-band. The EO/IR subsystem, using cameras, tracks drones within 3km, confirming activity at 2km. ETRI transferred this technology to Samjung Solution Co in 2022, leading to a 2024 public launch. Successfully tested in August 2024, this system neutralised unauthorised drones during national defence training.

AI predicts key metrics for sodium-ion batteries

Researchers at Tokyo University of Science predict sodium-ion batteries as a cost-effective alternative to lithium-ion, with a major challenge being the storage of larger sodium ions. They focused on transition-metal layered oxides, specifically NaMeO_2 , where 'Me' is a blend of metals like manganese, nickel, titanium, and iron. Using AI and 11 years of data from 68 compositions, the model predicted key battery metrics by optimising metal ratios for improved voltage and capacity. Their top result, $\text{Na}(\text{Mn}_{0.36}\text{Ni}_{0.44}\text{Ti}_{0.15}\text{Fe}_{0.05})\text{O}_2$, demonstrated high energy density, showcasing AI's potential in developing efficient sodium-ion batteries for sustainable energy.

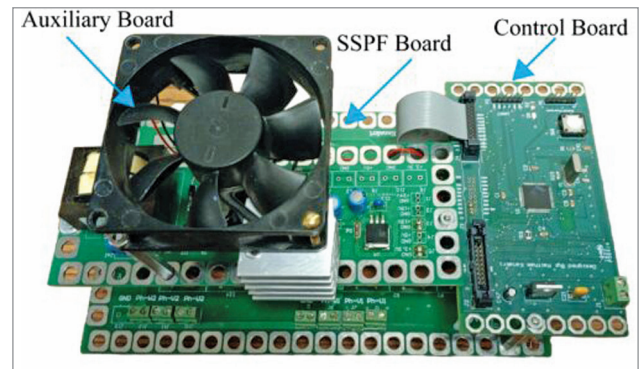
Memristor design makes computers smarter

Researchers at Wuhan University have created advanced memristors with high on/off ratios using two-dimensional van der Waals metallic materials like graphene and platinum ditelluride, achieving up to 10^8 on/off ratios with very low power consumption. Engineers are developing neuromorphic computing with memristors—components that mimic brain synapses by regulating currents and remembering past charges, potentially improving machine

learning efficiency. A CNN chip simulation using these memristors reached 91 % accuracy in image recognition, indicating strong potential for future AI applications and further material exploration.

Capacitorless filter could reform power electronics

A Purdue University research team has developed an innovative capacitorless solid-state power filter (SSPF) to enhance single-phase DC-AC converters. This SSPF replaces traditional LC filters and DC-link capacitors, utilising a



Experimental setup that shows: the inverter board part, SSPF board part, and the customised control board utilising dspic33ck256MP508 microcontroller (Credit: CES Transactions on Electrical Machines and Systems, 2024)

high-frequency planar transformer operating at 30kHz within an H-bridge converter, achieving a 1.29 % total harmonic distortion on a 120V, 60Hz system, compliant with IEEE 519 standards. The design reduces components, cost, and complexity, benefiting applications like EV chargers and renewable energy. Further, using calcium copper titanate (CCTO) increases stray capacitance, offering compact, efficient, capacitor-free DC-DC converters.

Graphene exhibits unique photoresistive properties

Researchers at the National University of Singapore have found that doped metallic graphene exhibits unique photoresistive properties when exposed to continuous-wave terahertz (THz) radiation. They observed that Dirac electrons in this material can thermally isolate from the lattice, creating a fluid-like, hydrodynamic flow. By designing graphene samples with narrow constrictions, the team achieved THz sensitivity as viscous effects altered the material's conductivity. This breakthrough led to developing viscous electron bolometers, which detect conductivity shifts rapidly. This research impacts next-gen wireless (6G +), autonomous vehicle navigation, and high-resolution astronomical imaging.